

HERNANDO, PANTINE B. CHALLENGE - 10 - CRYPTOGRAPHY

BSIT-3

rotation

Medium

Cryptography

picoCTF 2023

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Description

You will find the flag after decrypting this file
Download the encrypted flag [here](#).

Hints ?

1

29,070 users solved

89% Liked

picoCTF{FLAG}

Submit Flag

The **Rotation** challenge in picoCTF 2023 is a beginner-friendly cryptography task that involves decoding a message encrypted using a rotation cipher (also known as a Caesar cipher).

Objective

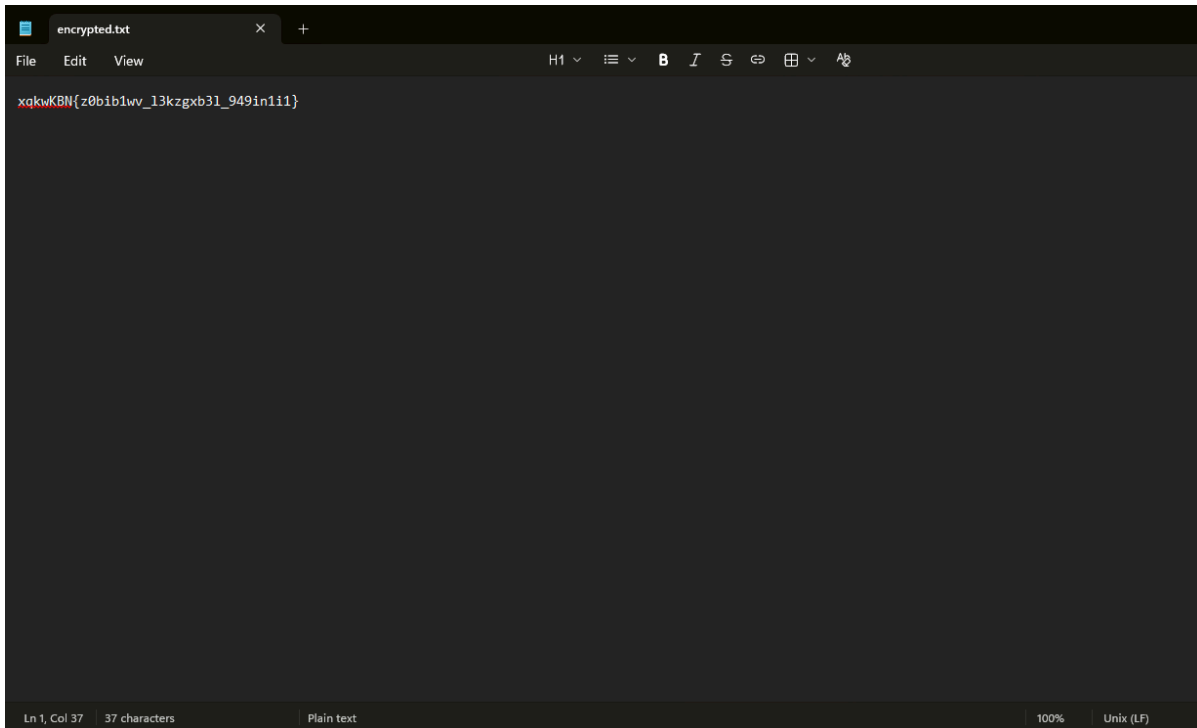
The goal is to decrypt the provided encoded file and retrieve the flag in the format:
`picoCTF{r0tat1on_d3crypt3d_949af1a1}`

Approach

Upon downloading the encrypted file, the contents appear as unreadable text. This suggests the use of a simple substitution cipher, specifically a rotation cipher where each letter is shifted by a fixed number in the alphabet.

Since the rotation value is unknown, the most effective method is to try all possible shifts (ROT1 to ROT25). This is considered a brute-force approach in basic cryptography.

Solution Process



```
xqkwKBW{z0bib1wy_13kzgb3l_949in111}
```

1. Open the encrypted file using a text editor.
2. Copy the ciphertext.
3. Use a rotation/caesar cipher tool (or script) to test all 25 possible shifts.
4. Scan the outputs for readable English text.
5. Identify the correct shift when the message becomes meaningful.
6. Locate the flag within the decoded text.

Key Insight

One of the rotations will clearly reveal a readable sentence containing the flag. The correct rotation stands out because it produces coherent words and follows the expected flag format.

Conclusion



The screenshot displays the dCode website's ROT Cipher decoder interface. At the top, a banner announces a new interface and encourages feedback on the 'ROT Cipher' page. The main section is titled 'ROT CIPHER' and 'ROT CIPHER DECODER'. It features a 'ROTATED TEXT' input field containing the ciphertext 'xqkwKBN{z0b1b1wv_13kzgx3l_949in11}'. Below this, there are two main decryption methods: 'AUTOMATIC DECRYPTION (BRUTE-FORCE)' and 'CUSTOM DECRYPTION'. The automatic method includes a dropdown for 'PLAINTTEXT LANGUAGE' set to 'English' and a 'DECRYPT' button. The custom method allows for selecting a rotation (N=13) and an alphabet to use, with options for Caesar, ROT13, and other ciphers. A 'DECRYPT' button is also present for the custom method. On the right side, there is a 'Summary' section with links to various cipher-related pages, a 'Similar pages' section with links to other ciphers, and a 'Support dCode' section with a link to the Discord forum. At the bottom, there is a 'Forum/Help' section with a Discord logo and a 'Keywords' section listing terms like 'rot', 'rotation', 'caesar', 'code', 'shift', 'rot13', 'rot47', 'alphabet', and 'circular'. A 'Links' section at the bottom right contains links to 'Contact', 'About dCode', and 'dCode App'.

Search for a tool

★ SEARCH A TOOL ON dCODE

e.g. type 'sudoku'

★ BROWSE THE FULL dCODE TOOLS' LIST

Results

↑↓	↑↓
[A-Z]+8	picoCTF{r0tat1on_d3crypt3d_949af1a1}
[A-Z0-9]+25	81v7VMY{abmtmc76_wevar8mew_kfkyctc}
[A-Z][0-9]+8	picoCTF{r2tat3on_d5crypt5d_161af3a3}
[A-Z0-9]+27	6zt5TKW{89krka54_uct8p6kcu_idirwara}
[A-Z0-9]+8	picoC3F{rs3a3ton_dvcr8p3vd_1w1afat}
[A-Z0-9]+23	a3x9X00{cdovoe98_ygxctaogy_mhmv0eve}
[A-Z0-9]+22	b4yaYP1{depwpfa9_zhydubphz_ninw1fwf}
[A-Z][0-9]+19	exrdRIU{g1pi1dc_s4rgnei4s_050pu2p2}
[A-Z]+19	exrdRIU{g0ipi1dc_s3rgnei3s_949pu1p1}
[A-Z0-9]+24	92w8WNZ{bcnund87_xfwbs9nfx_1gluzdud}
[A-Z]+10	ngamARD{p0ryr1m1_b3apwnr3b_949yd1y1}
ASCII[!-~]+5	s1frF=Ivu+]d],rqZg.fubs].gZ4/4di,d,x
ASCII[!-~]+40	PICO#x&SRf:A:g0N7DiCR?P:iD7ojoAFgAgU
[A-Z0-9]+29	4xr3RIU{67ipi832_sar6n4ias_gbgpu8p8}
ASCII[!-~]+8	picoC:Fsr(ZaZ)onWd+cr_pZ+dw1,1af)a)u
[A-Z0-9]+10	ngamA1D{pq181rm1_btap6n1tb_zuz8dr8r}

dCode is preparing a new interface. Come test and give your feedback on the new page: ROT Cipher!

ROT CIPHER

Cryptography · Substitution Cipher · ROT Cipher

ROT CIPHER DECODER

★ ROTATED TEXT

xqkwKBN{z0b1b1wv_13kzgx3l_949in11}

AUTOMATIC DECRYPTION (BRUTE-FORCE)

★ (EXPECTED) PLAINTEXT LANGUAGE: English

▶ DECRYPT

CUSTOM DECRYPTION

★ ROTATION TO USE ROT-N, N= 13

★ ALPHABET TO USE

☒ ABCDEFGHIJKLMNOPQRSTUVWXYZ (Rot13 / CAESAR)

☐ ABCDEFGHIJKLMNOPQRSTUVWXYZ AND 0123456789 (Rot13+5)

☐ ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789 (Rot18)

☐ 0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZ (Base36)

☐ 94 PRINTABLE ASCII CHARACTERS FROM !33 TO !126 (Rot47)

☐ FULL ASCII TABLE (128 CHARACTERS)

☐ CUSTOM ALPHABET (CASE INSENSITIVE)

1234567890QWERTYUIOPASDFGHJKLZXCVBNM

☐ CUSTOM ALPHABET (CASE SENSITIVE)

qwertyuiopasdfghjklzxcvbnm1234567890QWERTYUIOPASD...

▶ DECRYPT

See also: Caesar Cipher – Shift Cipher – ROT-13 Cipher – ROT-47 Cipher

ROT-N ENCODER

★ PLAIN TEXT TO ROTATE

dcode ROT

★ ROTATION TO USE ROT-N, N= 13

★ ALPHABET TO USE

☒ ABCDEFGHIJKLMNOPQRSTUVWXYZ (Rot13 / CAESAR)

☐ ABCDEFGHIJKLMNOPQRSTUVWXYZ AND 0123456789 (Rot13+5)

☐ ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789 (Rot18)

☐ 0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZ (Base36)

☐ 94 PRINTABLE ASCII CHARACTERS FROM !33 TO !126 (Rot47)

Summary

- ★ ROT Cipher Decoder
- ★ ROT-n Encoder
- ★ What is Rot cipher? (Definition)
- ★ How to encrypt using Rot cipher?
- ★ How to decrypt with Rot cipher?
- ★ What are rot variants?
- ★ Why is the ROT cipher used?
- ★ What is the Rot cipher algorithm?

Similar pages

- ★ Caesar Cipher
- ★ ROT-13 Cipher
- ★ ROT-47 Cipher
- ★ Shift Cipher
- ★ ROT1 Cipher
- ★ ROT-5 Cipher
- ★ ASCII Shift Cipher
- ★ dCODE'S TOOLS LIST

Support dCode

Participate in the Discord forum:

- ★ Discord

Forum/Help

DISCORD

Keywords

rot, rotation, caesar, code, shift, rot13, rot47, alphabet, circular

Links

- ★ Contact
- ★ About dCode
- ★ dCode App

This challenge demonstrates how weak simple encryption methods like rotation ciphers are, especially when brute-force techniques are applied. It also reinforces foundational skills in cryptography, such as pattern recognition and systematic testing.